Lecture 11

Algebras in tensor categories and module categories

Objectives. *A fusion category acts on a category by functors $X \otimes -$, one for each object X, with coherent associativity and unit isomorphisms. A category with such an action is a* module category. *This lecture builds module categories from algebras and computes a rank-one module associator. Ostrik's theorem says that every finite semisimple indecomposable module category comes from an algebra.*

By the end of this lecture you will be able to:

- Define algebras in C and their modules, and prove that the modules over an algebra form a C-module category.
- Write the module pentagon for a rank-one module category over the part of Vec_G^ω on a subgroup H, and show that it exists exactly when $\omega|_H$ is a coboundary.
- Compute internal Homs in a semisimple module category, and state Ostrik's theorem.
- Classify the indecomposable semisimple module categories over Vec_G by pairs (H, ψ), and list them for $\mathbb{Z}_2$.

11.1 Algebras and modules in C

Throughout, C is a fusion category, as in Lecture 5, and pictures suppress associators by coherence (Theorem 3.3).

Definition 11.1 (Algebra, module)**.** An *algebra* in C is (A, μ, ι) with $\mu\colon A \otimes A \to A$ and $\iota\colon 1 \to A$ such that

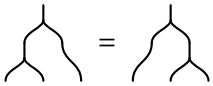
$$\begin{aligned} \mu \circ (\mu \otimes \mathrm{id}) &= \mu \circ (\mathrm{id} \otimes \mu) \circ \alpha_{A,A,A}, \\ \mu \circ (\iota \otimes \mathrm{id}) &= l_A, \\ \mu \circ (\mathrm{id} \otimes \iota) &= r_A \end{aligned}$$

(EGNO Def. 7.8.1, which writes m, u). A *right A-module* is (M, p) with $p\colon M \otimes A \to M$,

$$\begin{aligned} p \circ (p \otimes \mathrm{id}) &= p \circ (\mathrm{id} \otimes \mu) \circ \alpha_{M,A,A}, \\ p \circ (\mathrm{id} \otimes \iota) &= r_M. \end{aligned}$$

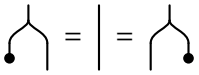
A module map is a morphism f of C with $f \circ p = p' \circ (f \otimes \mathrm{id})$; they form $\mathrm{Mod}_C(A)$ (EGNO Def. 7.8.5, Rem. 7.8.6). Left modules and *bimodules*, with commuting left and right actions, are defined alike.

With μ drawn as a fusion vertex and ι as a dot at the bottom of a strand, associativity is



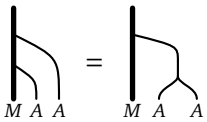
(11.1)

and the unit laws are



(11.2)

A right module is a thick strand, the *wall*, which A-strands join from the right; its axiom says that two A-strands may be multiplied before or after they join it:



(11.3)

Example 11.2 (Algebras)**.**

- In $\mathrm{Vect}_{\mathrm{fd}}$, an algebra is an ordinary finite-dimensional algebra, and $\mathrm{Mod}(A)$ its right modules.
- The unit 1, with $\mu = l_1$: its modules are all of C.
- For X in C, $X \otimes X^*$, with $\mu = \mathrm{id}_X \otimes \mathrm{ev}_X \otimes \mathrm{id}_{X^*}$ and $\iota = \mathrm{coev}_X$ (the duals of Lecture 4).
- In Vec_G^ω, for a subgroup $H \leq G$ and a function $\psi\colon H \times H \to \mathbb{C}^\times$: $A = \bigoplus_{h\in H} \delta_h$ with $\mu = \psi(g,h)$ on $\delta_g \otimes \delta_h \to \delta_{gh}$, the *twisted group algebra* $\mathbb{C}_\psi[H]$. Composing with $\alpha_{g,h,k} = \omega(g,h,k)$, associativity reads

$$\begin{aligned}&\psi(g,h)\,\psi(gh,k)\\ &= \omega(g,h,k)\,\psi(h,k)\,\psi(g,hk)\end{aligned}$$

  for $g,h,k \in H$, and the unit is δ_1 when $\psi(1,h) = \psi(h,1) = 1$.

For $\omega = 1$ the displayed equation says that ψ is a 2-*cocycle*; rescaling the basis of each δ_h by $\phi(h)$ changes ψ by the *coboundary* $(\delta\phi)(g,h) = \phi(g)\phi(h)/\phi(gh)$, and the classes form $H^2(H,\mathbb{C}^\times)$. These are exactly the cocycles of projective representations, $U_gU_h = \psi(g,h)U_{gh}$ (*Symmetries* Def. 13.19, which writes ω).

Every object X gives a *free* module $X \otimes A$, with $p = (\mathrm{id}\otimes\mu)\circ\alpha_{X,A,A}$, and module maps out of it are morphisms out of X: $\mathrm{Hom}_A(X\otimes A, M) \cong \mathrm{Hom}_C(X, M)$ (EGNO Lemma 7.8.12). An algebra is *separable* if μ has a right inverse $A \to A \otimes A$ that is a map of A-bimodules (EGNO Def. 7.8.29); then $\mathrm{Mod}_C(A)$ is semisimple (EGNO Prop. 7.8.30).

11.2 Module categories; the explicit module associator

Definition 11.3 (Module category)**.** A *left C-module category* is a $\mathbb{C}$-linear category $\mathcal{M}$ with a bilinear functor $\otimes\colon C \times \mathcal{M} \to \mathcal{M}$, a natural isomorphism, the *module associator*,

$$\begin{aligned}&\alpha^{\mathcal{M}}_{X,Y,M}\colon (X\otimes Y)\otimes M\\ &\to X\otimes(Y\otimes M),\end{aligned}$$

and a unit isomorphism $l^{\mathcal{M}}_M \colon 1 \otimes M \to M$, such that the *module pentagon*

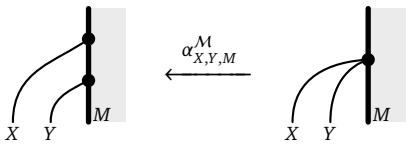
$$\begin{aligned}\alpha^{\mathcal{M}}_{X,Y,Z\otimes M} \circ \alpha^{\mathcal{M}}_{X\otimes Y,Z,M} \\ = (\mathrm{id}_X \otimes \alpha^{\mathcal{M}}_{Y,Z,M}) \circ \alpha^{\mathcal{M}}_{X,Y\otimes Z,M} \\ \circ (\alpha_{X,Y,Z} \otimes \mathrm{id}_M)\end{aligned}$$

and the triangle $(\mathrm{id}_X \otimes l^{\mathcal{M}}_M) \circ \alpha^{\mathcal{M}}_{X,1,M} = r_X \otimes \mathrm{id}_M$ hold (EGNO Defs. 7.1.1–7.1.2).

Example 11.4 (Module categories)**.** C acts on itself, with $\alpha^C = \alpha$: the *regular* module. A monoidal functor $(F, J, J_0) \colon C \to \mathrm{Vect}_{\mathrm{fd}}$, such as the forgetful functor of $\mathrm{Rep}(G)$, makes $\mathrm{Vect}_{\mathrm{fd}}$ a C-module category, with $X \otimes V = F(X) \otimes V$ and $\alpha^{\mathcal{M}}$ built from J^{-1} (EGNO Ex. 7.4.6).

A *module functor* $(F, s) \colon \mathcal{M} \to \mathcal{N}$ is a functor with natural isomorphisms $s_{X,M} \colon F(X \otimes M) \to X \otimes F(M)$ compatible with the associators and units (EGNO Def. 7.2.1). The module category $\mathcal{M}$ is *indecomposable* if it is not equivalent to a direct sum $\mathcal{M}_1 \oplus \mathcal{M}_2$ of two nonzero module categories (EGNO Def. 7.3.6).

In pictures $\mathcal{M}$ is a shaded region: an object M is a wall bounding it, and an object X of C acts by a strand that joins the wall from the unshaded side at a marked point. The module associator compares joining Y then X at two points with joining $X \otimes Y$ at one:



A module category is not a category of modules: its objects need not be modules over anything. Ostrik's theorem (§ 11.3) relates the two.

Theorem 11.5 (Modules form a module category · *proved given coherence*)**.** *For an algebra A in C, $\mathrm{Mod}_C(A)$ is a left C-module category: $X \otimes (M, p) = (X \otimes M, (\mathrm{id}_X \otimes p) \circ \alpha_{X,M,A})$, with module associator $\alpha_{X,Y,M}$ and unit l_M (EGNO Prop. 7.8.10).*

Proof. In pictures $X \otimes M$ is the strand X beside the wall M; A-strands join the wall from the right, past nothing, so the module axioms of $X \otimes M$ are those of M. The associator and unitors of C are module maps, by their naturality in the last variable and the pentagon of C; the module pentagon and triangle are the pentagon and triangle of C, with M as the fourth object. ∎

On a module category with one simple object M the invertible δ_h act by $h \otimes M \cong M$, and the module associator is a number.

Theorem 11.6 (A rank-one module category)**.** *Let ω be a normalised 3-cocycle on G, $H \leq G$, and C_H the full subcategory of Vec^{ω}_G on the δ_h, $h \in H$. Make $\mathrm{Vect}_{\mathrm{fd}}$ a C_H-module category with $\delta_h \otimes M = M$, $l^{\mathcal{M}} = \mathrm{id}$ and $\alpha^{\mathcal{M}}_{g,h,M} = \psi(g,h)\,\mathrm{id}_M$. The module pentagon and triangle hold if and only if $\psi(g,1) = \psi(1,g) = 1$ and, for $g, h, k \in H$,*

$$\begin{aligned}\psi(g,h)\,\psi(gh,k) \\ = \omega(g,h,k)\,\psi(h,k)\,\psi(g,hk),\end{aligned}$$

that is, $\omega|_H = (\delta\psi)^{-1}$ in the notation of Lecture 3. So such a module category exists exactly when $\omega|_H$ is a coboundary.

Proof. On simple objects every arrow is a scalar, as in Theorem 3.8. In the module pentagon for g,h,k the left side is $\psi(g,h)\,\psi(gh,k)$ and the right side $\psi(h,k)\,\psi(g,hk)\,\omega(g,h,k)$. The triangle for g gives $\psi(g,1)=1$, and the pentagon at $(1,1,k)$ then gives $\psi(1,k)=\psi(1,1)=1$. Conversely, if $\omega|_H=(\delta\psi)^{-1}$, the equation at $(1,1,k)$ and $(g,1,1)$ gives $\psi(1,k)=\psi(g,1)=\psi(1,1)$, and $\psi/\psi(1,1)$ is normalised and still solves it. ∎

The same equation gives associativity of $\mathbb{C}_\psi[H]$ in Example 11.2. Sheet 11 reconstructs this algebra. EGNO prints this condition as $\omega=d\psi$ (Ex. 7.8.3, (9.13)); its own coboundary (Ex. 1.7.1), associator ((2.19)) and module pentagon ((7.2)) give the one above.

For $\mathbb{Z}_2$ with $\omega(s,s,s)=-1$, the semion's category of Lecture 7, the equation at (s,s,s) reads $\psi(s,s)=-\psi(s,s)$: no module category $\mathrm{Vect}_{\mathrm{fd}}$ with s acting, and no fibre functor, as Lecture 8 stated (EGNO Ex. 5.1.3). For $H=G$ the module category $\mathrm{Vect}_{\mathrm{fd}}$ needs $[\omega]=0$ (Lecture 13).

Induced to all of G, the algebra $\mathbb{C}_\psi[H]$ in Vec_G^ω gives the semisimple module category $\mathrm{Mod}(\mathbb{C}_\psi[H])$, whose simple objects are the free modules $\delta_g\otimes\mathbb{C}_\psi[H]$, one for each coset gH, permuted by G · *cited* (EGNO Ex. 9.7.2, for right module categories, with the sign corrected as above). The coset is visible in $\mathrm{Hom}_A(\delta_g\otimes A,\delta_{g'}\otimes A)\cong\mathrm{Hom}(\delta_g,\delta_{g'}\otimes A)$, which is nonzero exactly when $g\in g'H$.

11.3 Internal Hom; Ostrik

An internal Hom reconstructs an algebra from a module category.

Definition 11.7 (Internal Hom)**.** For objects M,N of a C-module category $\mathcal{M}$, the *internal Hom* $[M,N]$ is an object of C with isomorphisms

$$\begin{aligned}&\mathrm{Hom}_C(X,[M,N])\\ &\cong\mathrm{Hom}_{\mathcal{M}}(X\otimes M,N),\end{aligned}$$

natural in X (EGNO §7.9, (7.21)). By Yoneda (Theorem 2.6) it is unique up to unique isomorphism.

Theorem 11.8 (The semisimple internal Hom)**.** *If $\mathcal{M}$ is semisimple with finitely many simple objects, then*

$$\begin{aligned}&[M,N]\\ &=\bigoplus_c\mathrm{Hom}_{\mathcal{M}}(c\otimes M,N)\otimes c,\end{aligned}$$

the sum over the simple objects c of C, where $V\otimes c$ means $c^{\oplus\dim V}$ with multiplicity space $V=\mathrm{Hom}_{\mathcal{M}}(c\otimes M,N)$, and no dual. The evaluation $\mathrm{ev}_{M,N}\colon[M,N]\otimes M\to N$ is, on each summand, the canonical map $\mathrm{Hom}_{\mathcal{M}}(c\otimes M,N)\otimes(c\otimes M)\to N$, and the composite of evaluations makes $[M,M]$ an algebra, with unit from $l_M^{\mathcal{M}}$.

Proof. For a simple d, Schur's lemma (Theorem 2.13) gives $\mathrm{Hom}(d,\bigoplus_c V_c\otimes c)=V_d$, which with $V_c=\mathrm{Hom}_{\mathcal{M}}(c\otimes M,N)$ is $\mathrm{Hom}_{\mathcal{M}}(d\otimes M,N)$; both sides are additive in X, and every X is a sum of simples. Under the adjunction, $\mathrm{id}_{[M,N]}$ corresponds to $\mathrm{ev}_{M,N}$, which is the stated map. The multiplication $[M,M]\otimes[M,M]\to[M,M]$ is the morphism corresponding to $\mathrm{ev}\circ(\mathrm{id}\otimes\mathrm{ev})\circ\alpha^{\mathcal{M}}$; associativity and the unit laws follow from the module pentagon and triangle, by the uniqueness in the adjunction. The unit corresponds to $l_M^{\mathcal{M}}\colon 1\otimes M\to M$. ∎

Example 11.9 (Internal Homs)**.** For the regular module, duality (Lecture 4) gives $\mathrm{Hom}(c \otimes x, y) \cong \mathrm{Hom}(c, y \otimes x^*)$, so $[x, y] = y \otimes x^*$ and $[1, 1] = 1$; in Vec_G, $[\delta_x, \delta_y] = \delta_{yx^{-1}}$.

For $G = \mathbb{Z}_2$ and $A = \mathbb{C}[\mathbb{Z}_2]$, the module category $\mathrm{Mod}(A)$ has one simple object $M = A$. Both $1 \otimes M$ and $s \otimes M$ are isomorphic to M, so both multiplicity spaces $\mathrm{Hom}(M, M)$ and $\mathrm{Hom}(s \otimes M, M)$ are $\mathbb{C}$. Hence $[M, M] = 1 \oplus s$. Fix a basis vector v_s of the line s. Let a_1 and a_s be the homogeneous basis of A. Choose $t_1 = \mathrm{id}_M$ and the evaluation map $t_s \colon s \otimes M \to M$ given by

$$t_s(v_s \otimes a_1) = a_s, \qquad t_s(v_s \otimes a_s) = a_1.$$

With $s \otimes s = 1$, two evaluations give

$$t_s \circ (\mathrm{id}_s \otimes t_s) = t_1.$$

Thus the product on $[M, M]$ reconstructs $\mathbb{C}[\mathbb{Z}_2]$. Sheet 11 checks the general product.

Theorem 11.10 (Ostrik · *cited*)**.** *Let $\mathcal{C}$ be a fusion category over $\mathbb{C}$ and $\mathcal{M}$ a semisimple indecomposable $\mathcal{C}$-module category with finitely many simple objects. For any nonzero M in $\mathcal{M}$, $A = [M, M]$ is a separable algebra, and $N \mapsto [M, N]$ is an equivalence of module categories $\mathcal{M} \simeq \mathrm{Mod}_{\mathcal{C}}(A)$.*

This is Ostrik's Theorem 1 (math/0111139, p. 10), who writes "semisimple" algebra; over $\mathbb{C}$ this is separable (Müger 2012, Lemma 3.14 and Thm. 3.21). Every finite semisimple module category is a direct sum of indecomposable ones (EGNO Prop. 7.6.7), so classifying them means classifying algebras up to equivalence of their module categories (*Morita equivalence* of algebras, EGNO Def. 7.8.17).

11.4 Vec_G: (H, ψ), worked for $\mathbb{Z}_2$; $\mathrm{Rep}(G)$

Theorem 11.11 (Module categories over Vec_G · *sketched given Ostrik's theorem*)**.** *Indecomposable semisimple Vec_G-module categories with finitely many simple objects, up to equivalence, correspond to pairs $(H, [\psi])$ with $H \leq G$ and $[\psi] \in H^2(H, \mathbb{C}^\times)$, up to conjugating both, through $(H, \psi) \mapsto \mathrm{Mod}(\mathbb{C}_\psi[H])$ (EGNO Ex. 7.4.10–7.4.11 and 9.7.2).*

Sketch. Each δ_g is invertible, so $\delta_g \otimes -$ permutes the simple objects, transitively since $\mathcal{M}$ is indecomposable. Fix a simple M and let H be its stabiliser. By Theorem 11.8, $A = [M, M] = \bigoplus_g \mathrm{Hom}(\delta_g \otimes M, M) \otimes \delta_g$ has a one-dimensional summand for each $h \in H$ and none elsewhere; choosing a basis vector in each gives $\mu = \psi(g, h)$, and associativity makes ψ a 2-cocycle (Example 11.2). Other basis vectors change ψ by a coboundary, and another simple $\delta_g \otimes M$ conjugates H and ψ. Ostrik's theorem gives $\mathcal{M} \simeq \mathrm{Mod}(A)$, and Theorem 11.5 gives a module category for every pair. ∎

Example 11.12 ($\mathbb{Z}_2$)**.** On $\mathbb{Z}_2$ a normalised 2-cocycle has one value, $\psi(s, s)$, and the cocycle condition at (s, s, s), $\psi(s, s)\psi(1, s) = \psi(s, s)\psi(s, 1)$, is automatic. Every value is a coboundary: $\psi = \delta\phi$ with $\phi(s)^2 = \psi(s, s)$. So $H^2(\mathbb{Z}_2, \mathbb{C}^\times) = 0$, and the subgroups 1 and $\mathbb{Z}_2$ give the two indecomposable module categories over Vec_G, $G = \mathbb{Z}_2$:

- $H = 1$, $A = \delta_1 = 1$: $\mathrm{Mod}(1) = \mathrm{Vec}_G$, the regular module, with two simple objects that s exchanges;
- $H = \mathbb{Z}_2$, $A = \mathbb{C}[\mathbb{Z}_2] = \delta_1 \oplus \delta_s$: one simple object $\delta_1 \otimes A \cong \delta_s \otimes A$, the module category $\mathrm{Vect}_{\mathrm{fd}}$ on which s acts trivially; it comes from the fibre functor that forgets the grading.

Over Vec_G^ω with $\omega(s,s,s) = -1$ only $H = 1$ survives, since $\omega|_{\mathbb{Z}_2}$ is not a coboundary (Theorem 11.6; EGNO Ex. 9.7.2): the regular module is the only one.

Lecture 13 meets these two as the two gapped boundaries of the toric code. Sheet 11 counts the four module categories over Vec_{S_3}.

Over $\mathrm{Rep}(G)$ the same labels classify: the indecomposable semisimple module categories are $\mathrm{Rep}^\psi(H)$, the ψ-projective representations of H, on which $\mathrm{Rep}(G)$ acts by restriction and tensor product $\cdot$ *cited* (Ostrik Thm. 2, p. 12; EGNO Cor. 7.12.20). The lists agree because $\mathrm{Rep}(G)$ is the dual of Vec_G with respect to $\mathrm{Vect}_{\mathrm{fd}}$ (Lecture 12) and module categories over a category and its dual correspond (EGNO Thm. 7.12.16).

Summary.

- *Handed on:* an algebra in C is (A, μ, ι); its right modules form a left C-module category $\mathrm{Mod}_C(A)$ (Definitions 11.1 and 11.3 and Theorem 11.5).
- *The module associator:* a rank-one module over the part of Vec_G^ω on H is a normalised ψ with $\psi(g,h)\psi(gh,k) = \omega(g,h,k)\psi(h,k)\psi(g,hk)$, which exists exactly when $\omega|_H$ is a coboundary (Theorem 11.6).
- *Ostrik:* in the semisimple case $[M,N] = \bigoplus_c \mathrm{Hom}(c \otimes M, N) \otimes c$, and every finite semisimple indecomposable module category is $\mathrm{Mod}([M,M])$ for any chosen nonzero object M (Theorems 11.8 and 11.10).
- Vec_G*:* indecomposable semisimple module categories are pairs $(H, [\psi])$ up to conjugacy; $\mathbb{Z}_2$ has two (Theorem 11.11 and Example 11.12).
- *Next:* Lecture 12 uses module categories to compare fusion categories (Morita equivalence) and builds the Drinfeld centre.

References

Etingof, Gelaki, Nikshych and Ostrik, *Tensor Categories*: cochains, Ex. 1.7.1, p. 8; no fibre functor on Vec_G^ω, Ex. 5.1.3, p. 91; module categories, Defs. 7.1.1–7.1.2, pp. 131–132; module functors, Def. 7.2.1, p. 134; indecomposable, Def. 7.3.6, p. 135; fibre functors as module structures, Ex. 7.4.6, p. 136; Vec_G and $\mathrm{Rep}(G)$, Ex. 7.4.9–7.4.11 and Rem. 7.4.12, pp. 136–137; direct sums, Prop. 7.6.7, p. 140; algebras and modules, Def. 7.8.1, Ex. 7.8.3, Def. 7.8.5, Rem. 7.8.6, Prop. 7.8.10 and Lemma 7.8.12, pp. 141–144; Morita equivalence of algebras, Def. 7.8.17, p. 145; separable algebras, Def. 7.8.29 and Prop. 7.8.30, p. 146; internal Hom, §7.9, pp. 147–150; Ostrik's theorem, Thm. 7.10.1, pp. 150–152; duals and module categories, Thm. 7.12.16, and $\mathrm{Rep}(G)$, Cor. 7.12.20, pp. 157–158; Vec_G^ω, Ex. 9.7.2 and Rem. 9.7.3, p. 287 (their $d\psi = \omega$ is our $\omega = (\delta\psi)^{-1}$). Ostrik, *Module categories, weak Hopf algebras and modular invariants*, math/0111139: module categories, Def. 6, pp. 5–6; internal Hom, Def. 11, p. 9; Theorem 1, §3.3, p. 10; $\mathrm{Rep}(G)$, Thm. 2, p. 12. Müger, *Modular categories* (2012): separable algebras, Def. 3.11 and Lemma 3.14, pp. 5–6, and Ostrik's theorem, Thm. 3.21, p. 6. Gruber, lecture notes: module categories (right-handed), Def. 2.1, pp. 9–10.